SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Deep Learning: Artificial Neural Networks, CNNs & Sequence Models

Department: CSE AIML	**Course / Subject:** Deep Learning (AI601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Perceptrons, Backpropagation, Convolutional Neural Networks (CNNs) for Computer Vision, Recurrent Neural Networks (RNNs, LSTMs, GRUs), and Attention Mechanisms.

Unit 1: Artificial Neural Networks (ANN)

Biological vs Artificial neurons, Perceptrons, Multilayer Perceptron (MLP), Activation functions (Sigmoid, Tanh, ReLU, LeakyReLU, Softmax), Backpropagation derivation.

Unit 2: Training Deep Networks

Vanishing and Exploding gradients, Weight initializations (He, Xavier), Optimizers (Momentum, RMSprop, Adam), Dropout regularization, Batch Normalization.

Unit 3: Convolutional Neural Networks (CNN)

Convolution operation, Padding, Strides, Pooling layers (Max, Average), Classic CNN architectures (LeNet, AlexNet, VGG-16, ResNet skip connections).

Unit 4: Recurrent Neural Networks (RNN)

Sequence modeling, Vanilla RNN architecture, Backpropagation Through Time (BPTT), Long Short-Term Memory (LSTM) cell gating, Gated Recurrent Units (GRU).

Unit 5: Transformers & Modern Architectures

Sequence-to-Sequence models, Self-Attention mechanism, Scaled Dot-Product attention, Multi-Head Attention, Transformer Encoder-Decoder, and Generative Adversarial Networks (GANs).